

MTH321 · NUMERICAL ANALYSIS OF PARTIAL DIFFERENTIAL EQUATIONS

Numerical Analysis of the Robertson Stiff Chemical Kinetics System

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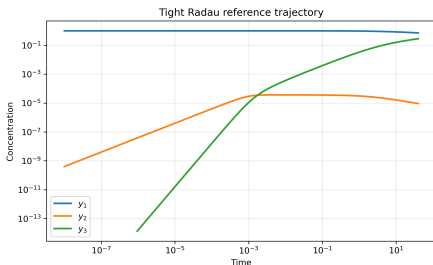
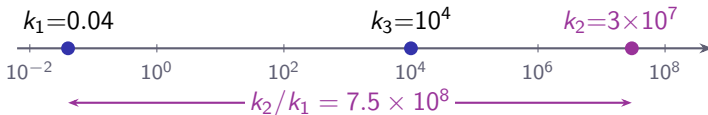
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Presentation Roadmap

- 1 **Motivation** *Why is this problem hard?*
- 2 **Model & Methods** *What do we solve, how?*
- 3 **Stability & Prediction** *Can we predict the limit?*
- 4 **Validation & Diagnostics** *Can we trust the numbers?*
- 5 **Efficiency & Insight** *Who is fastest, and why?*

What Makes This Problem Hard?



Radau reference: y_2 stays $\sim 10^{-5}$ throughout

Why a good testbed?

- 1 **No closed form**
needs a verified reference
- 2 **Exact invariant**, $y \geq 0$
checkable **independently**
of accuracy
- 3 **Predictable limit**
explicit step-size bound
from the Jacobian

Stiffness here = a stability restriction, not an accuracy restriction

The Robertson System

Mass-action ODE $(\mathbf{y} = (y_1, y_2, y_3)^\top = A, B, C; \mathbf{y}(0) = (1, 0, 0)^\top)$

$$\begin{aligned}y_1' &= -0.04 y_1 + 10^4 y_2 y_3, \\y_2' &= 0.04 y_1 - 10^4 y_2 y_3 - 3 \times 10^7 y_2^2, \\y_3' &= 3 \times 10^7 y_2^2\end{aligned}$$

$$\mathbf{1}^\top \mathbf{f}(\mathbf{y}) \equiv 0$$

$$\Rightarrow y_1 + y_2 + y_3 = 1$$

$\Rightarrow$ **structural zero
eigenvalue of J_f**

Full Jacobian J_f

spec $\{0, \lambda_1, \lambda_2\}$

naive $S = \infty$

exact

Reduced J_r (2×2)

spec $\{\lambda_1, \lambda_2\}$

via $y_3 = 1 - y_1 - y_2$

eig.

Stiffness ratio

$$S = \frac{\max_j |\operatorname{Re} \lambda_j|}{\min_j |\operatorname{Re} \lambda_j|}$$

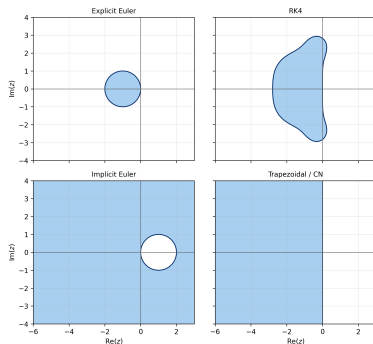
$S(0)$ is undefined ($y_2(0) = 0$), so all $S(t)$ curves start at $t = 10^{-8}$

Numerical Methods: Four Integrators

Method	$R(z)$	Order	Type	Stability
Explicit Euler	$1 + z$	1	explicit	bounded
RK4	$1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$	4	explicit	bounded
Implicit Euler	$\frac{1}{1 - z}$	1	implicit	A- and L-stable
Trapezoidal / CN	$\frac{1 + z/2}{1 - z/2}$	2	implicit	A-stable, not L

- Test equation $y' = \lambda y$, $z = h\lambda$; $y_{n+1} = R(z) y_n$
- **Implicit stages:** Newton–Raphson with the analytic 3×3 Jacobian, iteration matrix $I - hJ$ (CN: $I - \frac{h}{2}J$), stop at $\|g\|_\infty \leq \tau_N$
- **Adaptive:** step-doubling Explicit Euler and Implicit Euler (first-order / locally extrapolated $2y_{\text{two}} - y_{\text{full}}$)
- **Reference:** SciPy Radau, $\text{rtol} = 10^{-12}$, cross-checked with BDF

Stability Regions



$|R(z)| \leq 1$ (shaded), $\text{Re } z \in [-6, 3]$,

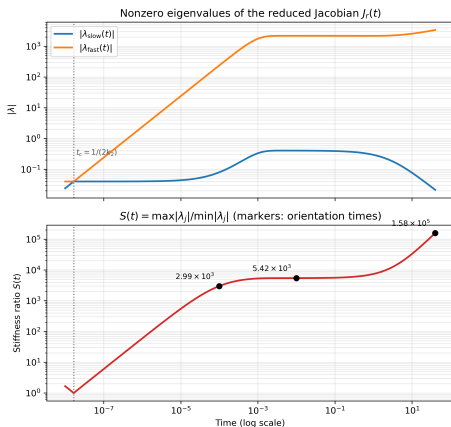
$\text{Im } z \in [-4, 4]$

- Real-axis intervals: Explicit Euler $[-2, 0]$; RK4 $[-2.7853, 0]$ ($\approx 1.4\times$ Euler, still bounded)
- Implicit Euler & CN: **whole left half-plane** (A-stable)
- Implicit Euler: $R(z) \rightarrow 0 \Rightarrow$ **L-stable**; CN: $R(z) \rightarrow -1 \Rightarrow$ **not** L-stable

Testable prediction

Very stiff modes are **damped** by Implicit Euler but **reflected** (amplitude ≈ 1 , alternating sign) by CN $\Rightarrow$ CN may lose non-negativity at large h .

Eigenvalue Analysis



$|\lambda_{\text{slow}}|$, $|\lambda_{\text{fast}}|$ and $S(t)$ along the Radau reference

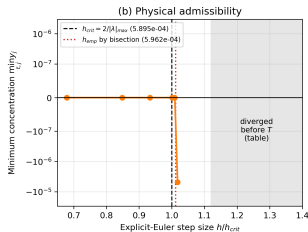
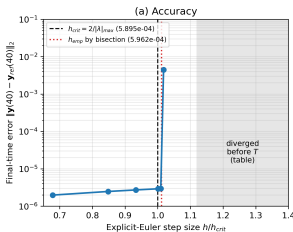
- $|\lambda_{\text{fast}}|$ saturates near 2.2×10^3 for $t \gtrsim 10^{-3}$, and reaches 3.393×10^3 at $t = 40$
- $S(t)$: plateau $\approx 5.4 \times 10^3$, then 1.58×10^5 at $t = 40$
- $S(t)$ matches the problem-pack values to $< 0.5\%$
- Non-normality is mild where it matters: $\kappa(V) \lesssim 2.2$ at late times

The fast rate is $O(10^3)$, not 3×10^7

Testable Step-Size Boundary

Prediction:
$$h_{\text{crit}} = \frac{2}{|\lambda|_{\text{max}}} = \frac{2}{3.393 \times 10^3} \approx 5.895 \times 10^{-4}$$

Explicit Euler	5.00×10^{-4}	5.95×10^{-4}	6.00×10^{-4}	7.00×10^{-4}	8.00×10^{-4}
h/h_{crit}	0.85	1.01	1.02	1.19	1.36
outcome	admissible	admissible	negative	diverged	diverged
$\min y_j / t_{\text{div}}$	0	0	-4.8×10^{-6}	$t \approx 27$	$t \approx 18$



Bisection: $h_{\text{emp}} \approx 5.962 \times 10^{-4}$, only **1.1%** above h_{crit}

Are the Numerical Results Trustworthy?

Reference checks

Radau vs. BDF

$$\approx 2 \times 10^{-11}$$

vs. $100\times$ looser Radau

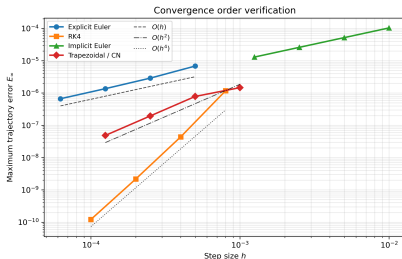
$$\approx 10^{-11}$$

Invariant defect

$$1.3 \times 10^{-15}$$

Fitted convergence orders

Method	formal	fitted
Explicit Euler	1	1.06
RK4	4	4.25
Implicit Euler	1	1.000
Trapezoidal / CN	2	2.004



Convergence of E_∞ against h

$$\mathbf{y}_{\text{ref}}(40) \approx (0.715827069, 9.1855 \times 10^{-6}, 0.284163746)$$

Diagnostics: Three Checks Fail Independently

Explicit Euler, $h = 6.0 \times 10^{-4}$

Diagnostic	value	verdict
invariant defect	6.7×10^{-15}	pass
$\min_j y_j$	-4.8×10^{-6}	fail
final error	4.5×10^{-3}	fail

Conservation holds by construction ($\mathbf{1}^\top \mathbf{f} \equiv 0$), so it says nothing about admissibility or accuracy.

Implicit Euler, Newton tol. ($h=10^{-2}$)

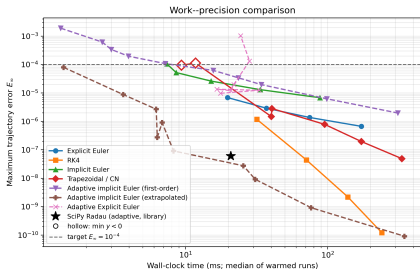
τ_N	status	$y_2(40)$
10^{-2}	non-physical	-3.6×10^{-6}
10^{-4}	failed ($t=3.71$)	—
10^{-6}	completed	9.191×10^{-6}
10^{-12}	completed	9.187×10^{-6}

τ_N must resolve $y_2 \sim 10^{-5}$, not the $O(1)$ components.

Monitor residual, y_2 , non-negativity **and** error — not conservation alone

A perturbed Newton initial guess passes the same residual test but converges to a non-physical root.

Efficiency

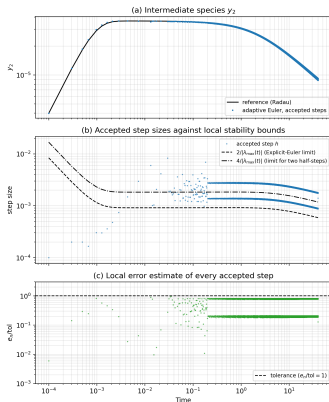


Method (h / tol)	E_∞	ms
IE, 5×10^{-3}	5.2×10^{-5}	8.6
RK4, 8×10^{-4}	1.2×10^{-6}	31.8
Adaptive EE, 10^{-7}	1.4×10^{-5}	16.6
Adaptive IE-ext, 10^{-5}	8.7×10^{-6}	3.6
Radau (library)	6.0×10^{-8}	20.8

median of warmed runs; machine-dependent

- At $\text{tol} = 10^{-6}$: extrapolated IE takes **501** steps (9.0×10^{-7}), adaptive EE **25 398** (2.0×10^{-5})
- Extrapolated IE is 0.8–22.5× faster than RK4/CN at matched error; RK4 edges ahead only at the tightest error tested

Why the Stability Restriction Persists



Adaptive EE, $\text{tol} = 10^{-6}$

Quasi-steady-state view

- $y_2^* \approx \sqrt{k_1 y_1 / k_2} \approx 3.65 \times 10^{-5}$
(0.08% from the reference peak)
- $\lambda_{\text{fast}} = -\sqrt{k_3^2 y_3^2 + 4k_1 k_2 y_1}$
- $h_{\text{crit}} = 2/|\lambda_{\text{fast}}| = 5.895 \times 10^{-4}$

What the step-doubling test sees

	Explicit Euler	Implicit Euler
fast mode in e_n	$z^2/4$ (grows)	≤ 0.09 (bounded)
step set by	fast mode	slow dynamics

$R_{\text{CN}} \rightarrow -1$ likewise explains negative CN values

Adaptivity changes h , but does **not** lift the explicit stability limit

What We Learned

1. **Results are trustworthy:** orders 1.06, 4.25, 1.000, 2.004; reference, invariant and step-size prediction all agree
2. **Stiffness = stability restriction, not accuracy:**
 $h_{\text{crit}} = 5.895 \times 10^{-4}$ vs. $h_{\text{emp}} = 5.962 \times 10^{-4}$; adaptivity does not lift it for explicit methods
3. **L-stable implicit methods succeed:** IE reaches $E_{\infty} \approx 10^{-4}$ at $h = 10^{-2}$ ($\approx 17 h_{\text{crit}}$); CN is not L-stable and loses non-negativity
4. **Keep diagnostics apart:** conservation, non-negativity, Newton residual and error fail independently

Practical recommendation

Use an L-stable implicit or stiff adaptive solver; choose $\tau_N \ll y_2$; monitor non-negativity and error, not just conservation.

Thank you — Questions?